

UC SAN DIEGO / ECE 251C

4

Sampling and oversampling

Sinc reconstruction, noisy samples, and mean-squared error

How does the sampling rate control the error caused by noisy samples?

The sampling theorem tells us when exact samples determine a bandlimited signal. In practice, the sample values are not exact. We will keep the same signal model, add independent noise to the samples, and calculate the resulting reconstruction error. The main tools are the Poisson summation formula and Parseval's identity.

4.1 Bandlimited signals and the sampling theorem

Definition. A signal $f \in L^2(\mathbb{R})$ is *bandlimited to* $B > 0$ if

$$\text{supp } \hat{f} \subseteq [-B, B], \quad \text{that is,} \quad \hat{f}(\omega) = 0 \quad \text{for } |\omega| > B.$$

Here B is an *angular* frequency. We sample the continuous version of f supplied by the inverse Fourier transform.

Theorem 4.1 (Sampling theorem). Let $f \in L^2(\mathbb{R})$ be bandlimited to B . If $0 < T < \pi/B$, then the samples $f[n] := f(nT)$ determine f uniquely, and

$$f(t) = \sum_{n \in \mathbb{Z}} f[n] \text{sinc}\left(\frac{t - nT}{T}\right), \quad (4.1)$$

where $\text{sinc}(s) = \sin(\pi s)/(\pi s)$ for $s \neq 0$ and $\text{sinc}(0) = 1$.

This is the Whittaker–Nyquist–Kotelnikov–Shannon sampling theorem. We recall its spectral argument on the next page.

Why “more than twice the highest frequency”? For a time signal, write $\omega = 2\pi\nu$, with ν measured in hertz. Set $\widetilde{B} = B/(2\pi)$. Then

$$T < \frac{\pi}{B} \iff T < \frac{1}{2\widetilde{B}} \iff \underbrace{\frac{1}{T}}_{\text{samples per second}} > 2 \underbrace{\widetilde{B}}_{\text{highest frequency in Hz}}. \quad (4.2)$$

The factors of 2π convert units; they do not change the sampling requirement.

4.2 Recall: sampling repeats the spectrum

Our Fourier convention is

$$\hat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-j\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\omega) e^{j\omega t} d\omega.$$

The sampled impulse train is

$$f_T(t) = f(t) \text{III}_T(t) = \sum_{n \in \mathbb{Z}} f[n] \delta(t - nT), \quad \omega_0 = \frac{2\pi}{T}.$$

Recall (Poisson summation). The identity from Lecture 3 is

$$\hat{f}_T(\omega) = \sum_{n \in \mathbb{Z}} f[n] e^{-j\omega n T} = \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{f}(\omega - k\omega_0). \quad (4.3)$$

The copies of $\hat{f}$ are separated by ω_0 . When $B < \omega_0/2$, only the central copy lies in $[-\pi/T, \pi/T]$.

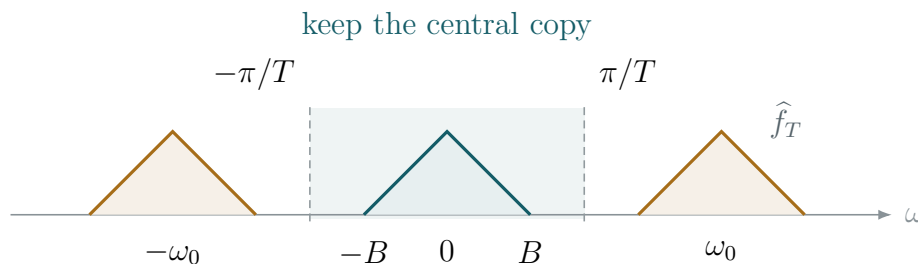


Figure 4.1: The shaded interval contains the original spectral copy and no others. Its amplitude is $\hat{f}/T$, so reconstruction must also restore the factor T .

Proof of the reconstruction formula. Choose

$$\hat{h}_T(\omega) = T \mathbf{1}_{[-\pi/T, \pi/T]}(\omega).$$

Then $\hat{h}_T \hat{f}_T = \hat{f}$. The inverse transform of this rectangle is, for $t \neq 0$,

$$\begin{aligned} h_T(t) &= \frac{T}{2\pi} \int_{-\pi/T}^{\pi/T} e^{j\omega t} d\omega \\ &= \frac{T}{2\pi} \frac{e^{j\pi t/T} - e^{-j\pi t/T}}{jt} = \text{sinc}(t/T). \end{aligned}$$

At $t = 0$, the integral equals one. Convolution with f_T gives

$$(h_T * f_T)(t) = \sum_{n \in \mathbb{Z}} f[n] h_T(t - nT) = f(t).$$

This proves [equation \(4.1\)](#) and establishes uniqueness. □

4.3 Oversampling and noisy measurements

Setup. Keep the bandlimit B fixed and choose a sampling step satisfying

$$\omega_0 = \frac{2\pi}{T} \geq 2B, \quad r := \frac{\omega_0}{2B} = \frac{\pi}{BT} \geq 1. \quad (4.4)$$

The number r is the *oversampling factor*. At $r = 1$ the allowed spectral bands just meet; at $r > 1$ they are separated by a gap. Equality is allowed in this finite-energy calculation: the values at the band endpoints do not affect any Fourier integral.

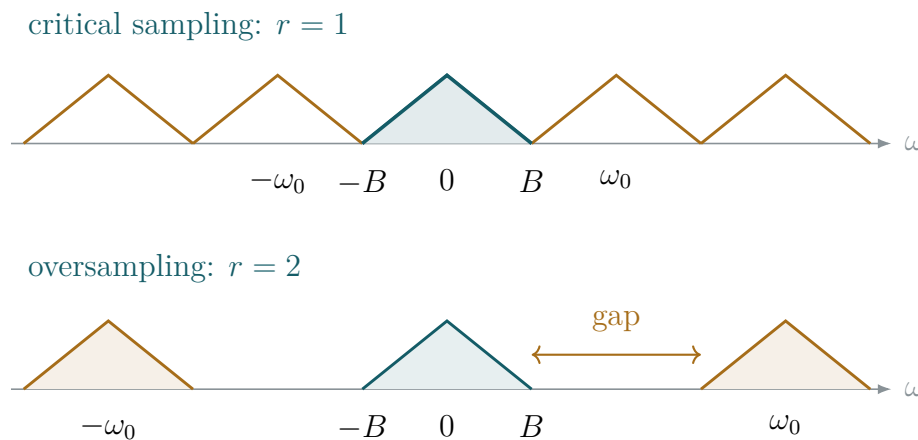


Figure 4.2: The signal band stays fixed while the spectral copies move apart. Both rows show $T\hat{f}_T$, so their heights are comparable even though the sampling steps differ.

Measurements. Instead of the exact samples, we observe

$$y[n] = f[n] + w[n], \quad \mathbb{E}\{w[n]\} = 0, \quad \mathbb{E}\{|w[n]|^2\} = \sigma^2. \quad (4.5)$$

Assume the $w[n]$ are independent and identically distributed. If $n \neq m$, independence and zero mean give

$$\mathbb{E}\{w[n]w^*[m]\} = \mathbb{E}\{w[n]\} \mathbb{E}\{w^*[m]\} = 0.$$

If $n = m$, the expectation is σ^2 . In one formula,

$$\mathbb{E}\{w[n]w^*[m]\} = \sigma^2 \delta[n - m]. \quad (4.6)$$

Only these second-moment properties enter the calculation; no Gaussian assumption is needed. We keep σ^2 fixed as the sampling rate changes.

4.4 Reconstruct only the signal band

For exact reconstruction, we do not have to keep every frequency up to π/T . We know that f has no frequencies outside $[-B, B]$. Choose the *signal-band* low-pass response

$$\hat{h}_{\text{LP}}(\omega) = T \mathbf{1}_{[-B, B]}(\omega). \quad (4.7)$$

It still satisfies $\hat{h}_{\text{LP}} \hat{f}_T = \hat{f}$: it keeps the entire central copy, removes the others, and cancels the sampling factor $1/T$.

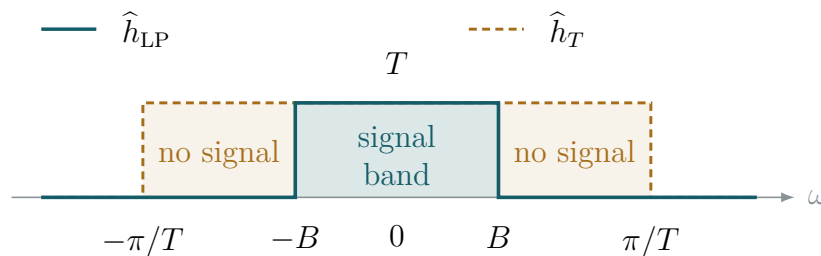


Figure 4.3: Two filters reconstruct the same noiseless signal. The solid response rejects the extra frequency bands admitted by the dashed, full-Nyquist response. With noisy samples, that distinction matters.

Calculate the impulse response. Inverting [equation \(4.7\)](#), for $t \neq 0$,

$$\begin{aligned} h_{\text{LP}}(t) &= \frac{1}{2\pi} \int_{-B}^B T e^{j\omega t} d\omega \\ &= \frac{T}{2\pi} \left[\frac{e^{j\omega t}}{jt} \right]_{-B}^B = \frac{T}{2\pi} \frac{2j \sin(Bt)}{jt} \\ &= \frac{T \sin(Bt)}{\pi t} = \frac{BT}{\pi} \text{sinc}\left(\frac{Bt}{\pi}\right). \end{aligned} \quad (4.8)$$

At zero, the integral gives $h_{\text{LP}}(0) = BT/\pi = 1/r$. Thus

$$h_{\text{LP}}(t) = \frac{1}{r} \text{sinc}\left(\frac{t}{rT}\right).$$

At $r = 1$ this is $h_T(t) = \text{sinc}(t/T)$. When $r > 1$, the two kernels differ. The signal-band kernel need not vanish at other sample locations: exact reconstruction comes from combining the samples, not from each sample acting in isolation.

4.5 The reconstruction error is filtered noise

Reconstruct from the measured values using the same low-pass kernel:

$$\tilde{f}(t) := \sum_{n \in \mathbb{Z}} y[n] h_{\text{LP}}(t - nT). \quad (4.9)$$

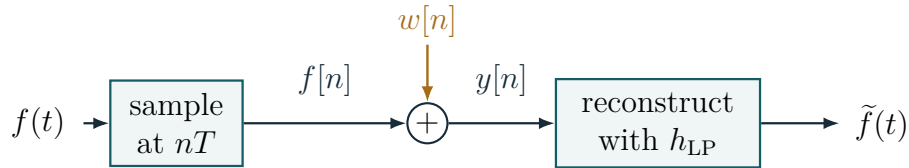


Figure 4.4: Noise is added to the samples. The reconstruction filter acts on both the signal samples and the noise samples.

Substitute $y[n] = f[n] + w[n]$ and use exact noiseless reconstruction:

$$\begin{aligned} \tilde{f}(t) &= \sum_n (f[n] + w[n]) h_{\text{LP}}(t - nT) \\ &= \underbrace{\sum_n f[n] h_{\text{LP}}(t - nT)}_{f(t)} + \sum_n w[n] h_{\text{LP}}(t - nT). \end{aligned}$$

Consequently the error is

$$e(t) := \tilde{f}(t) - f(t) = \sum_{n \in \mathbb{Z}} w[n] h_{\text{LP}}(t - nT). \quad (4.10)$$

Calculate its mean-squared size. Use $|z|^2 = zz^*$ and linearity of expectation:

$$\begin{aligned} \mathbb{E}\{|e(t)|^2\} &= \mathbb{E}\left\{\left|\sum_{n \in \mathbb{Z}} w[n] h_{\text{LP}}(t - nT)\right|^2\right\} \\ &= \sum_{n, m \in \mathbb{Z}} \mathbb{E}\{w[n] w^*[m]\} h_{\text{LP}}(t - nT) h_{\text{LP}}^*(t - mT). \end{aligned}$$

By [equation \(4.6\)](#), the terms with $n \neq m$ vanish. Each surviving term has $n = m$ and contributes $\sigma^2 |h_{\text{LP}}(t - nT)|^2$. Therefore

$$\mathbb{E}\{|e(t)|^2\} = \sigma^2 \sum_{n \in \mathbb{Z}} |h_{\text{LP}}(t - nT)|^2. \quad (4.11)$$

We will evaluate this sum using Poisson summation and Parseval's identity.

4.6 A reflected and shifted reconstruction kernel

Fix the location t at which we want to evaluate the error. Define

$$a_t[n] := h_{\text{LP}}(t - nT), \quad g_t(u) := h_{\text{LP}}(t - u). \quad (4.12)$$

Then $a_t[n] = g_t(nT)$: the weights are samples of the function g_t . The subscript t is a fixed parameter, while u is the variable being sampled or integrated.

Calculate the Fourier transform of g_t . Set

$$v = t - u, \quad u = t - v, \quad du = -dv.$$

The integration limits change as well:

$$u \rightarrow -\infty \implies v \rightarrow +\infty, \quad u \rightarrow +\infty \implies v \rightarrow -\infty.$$

Writing both changes explicitly gives

$$\begin{aligned} \hat{g}_t(\omega) &= \int_{-\infty}^{\infty} h_{\text{LP}}(t - u) e^{-j\omega u} du \\ &= - \int_{+\infty}^{-\infty} h_{\text{LP}}(v) e^{-j\omega(t-v)} dv \\ &= \int_{-\infty}^{\infty} h_{\text{LP}}(v) e^{-j\omega(t-v)} dv \\ &= e^{-j\omega t} \int_{-\infty}^{\infty} h_{\text{LP}}(v) e^{j\omega v} dv \\ &= e^{-j\omega t} \hat{h}_{\text{LP}}(-\omega). \end{aligned} \quad (4.13)$$

The minus sign from $du = -dv$ cancels the minus sign from reversing the limits. The positive exponential inside the last integral changes the *frequency argument* to $-\omega$; it does not introduce a minus sign in front of the transform.

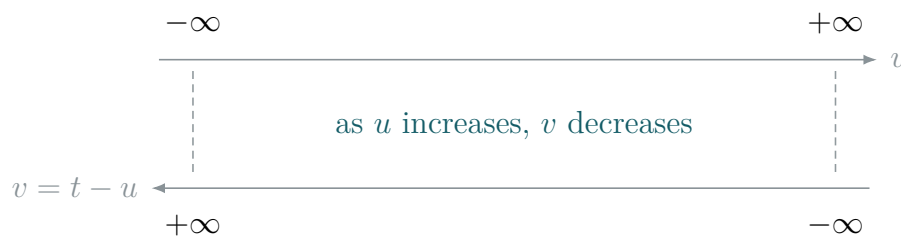


Figure 4.5: The substitution reverses orientation. Keeping the original integration limits while inserting only $du = -dv$ would create an incorrect extra minus sign.

4.7 Apply Poisson summation to the weights

Use physical angular frequency to write the DTFT of a_t :

$$A_t(\omega) := \sum_{n \in \mathbb{Z}} a_t[n] e^{-j\omega n T}. \quad (4.14)$$

It is ω_0 -periodic, since $e^{-j\omega_0 n T} = e^{-j2\pi n} = 1$.

Recall (Poisson summation). For a bandlimited finite-energy function g with samples $g[n] = g(nT)$,

$$\sum_{n \in \mathbb{Z}} g[n] e^{-j\omega n T} = \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{g}(\omega - k\omega_0), \quad \omega_0 = \frac{2\pi}{T}. \quad (4.15)$$

Apply this identity to the bandlimited function $g = g_t$ from [equation \(4.12\)](#). Substituting [equation \(4.13\)](#) gives

$$\begin{aligned} A_t(\omega) &= \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{g}_t(\omega - k\omega_0) \\ &= \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{h}_{\text{LP}}(-\omega + k\omega_0) e^{-j(\omega - k\omega_0)t}. \end{aligned} \quad (4.16)$$

Which terms survive on one period? If $|\omega| < \omega_0/2$ and $k \neq 0$, then

$$|-\omega + k\omega_0| \geq |k|\omega_0 - |\omega| > \frac{\omega_0}{2} \geq B.$$

The corresponding filter value is zero. Only $k = 0$ remains, so on the base interval

$$A_t(\omega) = \frac{1}{T} \hat{h}_{\text{LP}}(-\omega) e^{-j\omega t} = \mathbf{1}_{[-B, B]}(\omega) e^{-j\omega t}. \quad (4.17)$$

Here $\hat{h}_{\text{LP}}(-\omega) = T \mathbf{1}_{[-B, B]}(\omega)$ because the passband is symmetric. Endpoints do not affect the energy calculation.

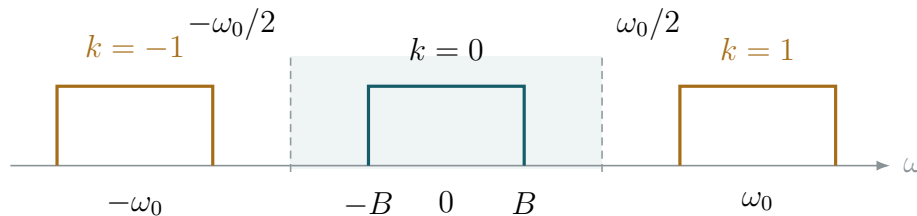


Figure 4.6: Magnitudes of three terms in the periodization. The shaded base interval intersects only the central rectangle. The phase factor depends on t , but its magnitude is always one.

4.8 Parseval: track the frequency normalization

The ordinary DTFT uses a dimensionless angular frequency θ :

$$\sum_{n \in \mathbb{Z}} a_t[n] e^{-j\theta n} = A_t(\theta/T), \quad \theta = \omega T.$$

Recall (Parseval). Sequence energy equals the normalized integral of squared DTFT magnitude over one period:

$$\sum_{n \in \mathbb{Z}} |a_t[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |A_t(\theta/T)|^2 d\theta. \quad (4.18)$$

Substitute $\theta = \omega T$, with $d\theta = T d\omega$. Since $\omega_0 = 2\pi/T$, the new integration limits are $-\omega_0/2$ and $\omega_0/2$. Hence

$$\begin{aligned} \sum_n |a_t[n]|^2 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |A_t(\theta/T)|^2 d\theta \\ &= \frac{T}{2\pi} \int_{-\omega_0/2}^{\omega_0/2} |A_t(\omega)|^2 d\omega \\ &= \frac{T}{2\pi} \int_{-B}^B 1 d\omega = \frac{BT}{\pi} = \frac{1}{r}. \end{aligned} \quad (4.19)$$

The third line uses [equation \(4.17\)](#): $|e^{-j\omega t}|^2 = 1$. Therefore the answer is independent of the reconstruction location t .

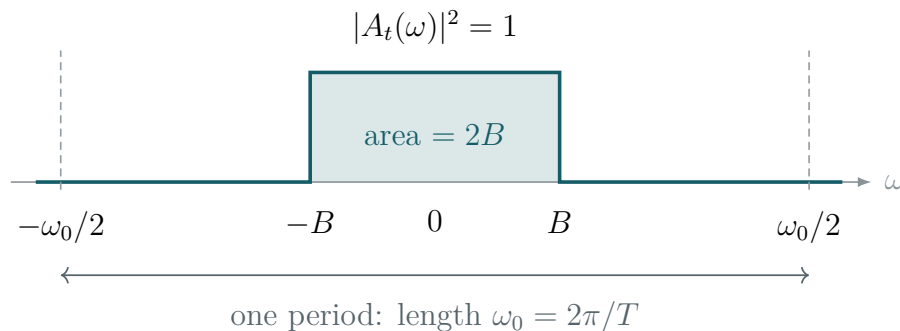


Figure 4.7: Parseval averages this squared magnitude over a period. The occupied fraction is $2B/\omega_0 = BT/\pi = 1/r$. Increasing the sampling rate enlarges the period, not the signal band.

4.9 The reconstruction error falls as $1/r$

Combining equations (4.11) and (4.19) gives the exact error variance:

$$\mathbb{E}\{|\tilde{f}(t) - f(t)|^2\} = \sigma^2 \frac{BT}{\pi} = \frac{\sigma^2}{r}, \quad t \in \mathbb{R}. \quad (4.20)$$

The error has mean zero because the sample noise has mean zero. Its mean-squared size is therefore also its variance. The expectation averages over noise realizations at a *fixed location*; it is not an integral of the error over the whole line.

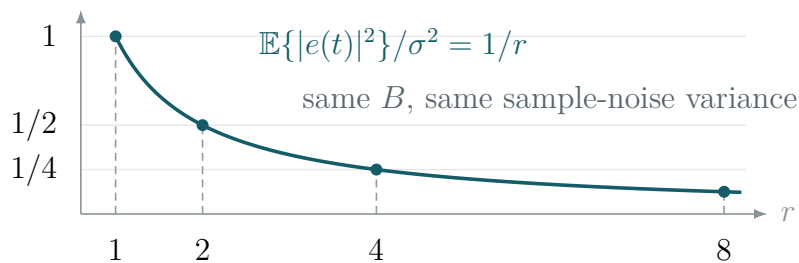


Figure 4.8: The exact normalized mean-squared error, not an empirical fit. Doubling the oversampling factor halves the variance. The root-mean-squared error decreases by a factor of $\sqrt{2}$.

Example. Suppose the per-sample noise standard deviation is $\sigma = 0.1$. Then $\sigma^2 = 0.01$. At $r = 2$, the reconstructed error variance is 0.005; at $r = 4$, it is 0.0025, with standard deviation 0.05.

Why the filter matters. As a worked consequence, suppose instead that we reconstruct using the full-Nyquist kernel $h_T(t) = \text{sinc}(t/T)$. Repeating the same calculation with cutoff π/T gives

$$\sum_n |h_T(t - nT)|^2 = \frac{T}{2\pi} \int_{-\pi/T}^{\pi/T} 1 \, d\omega = 1.$$

Its error variance is σ^2 , not σ^2/r . It reconstructs the noiseless signal too, but retains noise in frequencies where the signal is known to be zero.

What oversampling buys

With the signal band and per-sample noise variance fixed, more samples allow the signal-band reconstruction filter to reject a larger fraction of the noise. The improvement is a mean-squared-error statement under the specified independent-noise model, not a guarantee for every individual noise realization.